Finite quotients preserving spherical residues

Human Compression project

Abstract

For a Coxeter system with finite maximal parabolics, a finite set of products of three parabolics detects every failure of flag lifting in a quotient. Avoiding this set preserves the entire incidence structure of each proper residue, as well as the intersection property. Reduction of the faithful geometric representation modulo a suitable prime produces such quotients of arbitrarily large order whenever the Coxeter group is infinite. This gives a common proof of the finite existence assertions in Section 6 of the fixed manuscript, including the diagrams which are not strings. It does not determine the smallest quotient or verify the groups in the source's finite table.

1 A finite obstruction set

Let (W, S) be a Coxeter system of rank $r \geq 3$, with $S = \{s_i : i \in I\}$. Put $W_J = \langle s_j : j \in J \rangle$ for $J \subseteq I$, and write $P_i = W_{I\setminus\{i\}}$. Assume that every P_i is finite. We use the standard Coxeter complex: its objects of type i are the left cosets wP_i, and two objects of different types are incident when their cosets intersect. We retain the classical Coxeter facts that $W_J \cap W_K = W_{J\cap K}$, that this incidence system is a thin, residually connected geometry, and that W acts regularly on its chambers. In particular every pairwise incident set extends to a chamber. These are inputs about the universal complex, not assumptions about a proposed quotient.

Define the finite set

$$\mathcal{F} = \left(\bigcup_{i,j,k\in I} P_i P_j P_k \right) \setminus \{1\}.$$

Repetitions among i, j, k are allowed. Since every subgroup contains the identity, this set also contains all nonidentity elements in any P_i or $P_i P_j$.

Theorem 1. *Let $N \triangleleft W$ satisfy $N \cap \mathcal{F} = \varnothing$, and let $q : W \to Q = W/N$. The marked group $(Q, (q(s_i))_{i\in I})$ has the full intersection property. Its coset incidence system with maximal subgroups $q(P_i)$ is a regular hypertope. The residue of its base object of type i is isomorphic, as a typed incidence geometry, to the residue of P_i in the universal Coxeter complex. Thus every proper residue remains universal and spherical. The quotient map preserves every bond label.*

Proof. First, q is injective on every P_i. More generally, let J, K be proper subsets of I and choose $i \notin J$, $j \notin K$. If $a \in W_J$, $b \in W_K$ and $q(a) = q(b)$, then $ab^{-1} \in N \cap P_i P_j$, so $a = b$. The intersection property in W therefore gives

$$q(W_J) \cap q(W_K) = q(W_{J\cap K}).$$

When either subset is all of I, this equality is immediate. This proves the full intersection property, including the trivial intersection of all maximal marked parabolics. Every subgroup generated by two marks is contained in a maximal parabolic, since $r \geq 3$. Its injection proves that no mark disappears and that all pairwise product orders are unchanged.

We next prove the required flag lifting, rather than inferring it from these subgroup equalities. Let a nonempty flag in the quotient contain an object of type i. Translate it so that this object is $q(P_i)$. Every other object, of type j, is incident with $q(P_i)$, and hence has a representative

$$q(a_j)q(P_j), \qquad a_j \in P_i.$$

Choose such a representative separately for every type in the flag. The corresponding objects a_jP_j are all incident with P_i. For distinct types j, k in the flag, quotient incidence says

$$q(a_j^{-1}a_k) \in q(P_jP_k).$$

Choose $b \in P_j$, $c \in P_k$ with $q(a_j^{-1}a_k) = q(bc)$. Then

$$a_j^{-1}a_kc^{-1}b^{-1} \in N \cap P_iP_kP_j.$$

The avoidance hypothesis makes this element the identity. Consequently $a_j^{-1}a_k = bc \in P_jP_k$, which is exactly the incidence of a_jP_j and a_kP_k. Thus the chosen objects form an actual flag upstairs. They extend to a chamber of the Coxeter complex. Its image is a chamber containing the prescribed quotient flag. The same argument proves that every quotient flag is a translate of a subflag of the base chamber: lift, extend and use the chamber action of W. In particular Q is flag transitive.

Here is the asserted residue isomorphism in detail. An object of type $j \neq i$ incident with P_i has the form aP_j for $a \in P_i$. Mapping it to $q(a)q(P_j)$ is surjective onto the corresponding quotient residue. If $a, b \in P_i$ give the same image object, then $q(a^{-1}b) \in q(P_j)$. The already proved intersection equality and injectivity on P_i imply $a^{-1}b \in P_i \cap P_j$, so the objects upstairs coincide. This proves injectivity on objects. The preceding triple-product calculation proves that incidence of any two image objects reflects incidence upstairs. Incidence is preserved automatically. Hence the map is an isomorphism of the complete residue geometries, not merely of their groups or their chamber sets.

Every nonempty proper residue in the quotient is therefore a residue of a finite spherical Coxeter complex. It is thin and, when its rank is at least two, connected. The whole quotient geometry is connected as well: its chambers are the translates of the base chamber, with trivial chamber stabilizer by the intersection property. Multiplying successively by the marks gives galleries joining these chambers, because the marks generate Q. A gallery gives connectedness of the incidence graph in rank at least two. Thinness and residual connectedness now hold in all required ranks. Finally, in this geometry, a type-preserving automorphism fixing a chamber fixes its adjacent chambers and then every chamber, by the gallery connectedness just proved. Thus the full type-preserving automorphism group acts freely on chambers. Since Q already acts transitively and freely, it is that full group. □

The third factor has a specific role: two objects represented inside one fixed parabolic may become incident after quotienting. Pairwise parabolic injection alone does not rule this out. The argument does not claim that the displayed avoidance condition is necessary, or that it is economical for finding a small quotient.

2 Separating finitely many matrices

We give the elementary finite-quotient argument needed here. It also specifies what is retained from Coxeter theory.

Lemma 2. *Let K be a number field with ring of integers $\mathcal{O}_K$, and let $G \leq \mathrm{GL}_d(\mathcal{O}_K)$. For every finite set $A \subset G \setminus \{1\}$ there is a homomorphism from G to a finite group whose kernel is disjoint from A.*

Proof. For every $g \in A$, choose a nonzero entry a_g of $g - I$. Its algebraic norm $\mathrm{Norm}_{K/\mathbb{Q}}(a_g)$ is a nonzero integer: it is the determinant of multiplication by a_g on any integral basis of $\mathcal{O}_K$. Choose a rational prime p not dividing the product of these finitely many integers; for empty A any prime will do. Modulo p, each multiplication matrix remains invertible. In particular a_g has nonzero image in the finite ring $\mathcal{O}_K/p\mathcal{O}_K$.

Reduction of matrices is a homomorphism

$$G \longrightarrow \mathrm{GL}_d(\mathcal{O}_K/p\mathcal{O}_K).$$

Inverses remain inverses after reduction. The target is finite, since the additive group of the coefficient ring has $p^{[K:\mathbb{Q}]}$ elements. The chosen entry of every $g - I$ survives, so none of the elements of A maps to the identity. □

For a finite-rank Coxeter system the standard geometric representation has, in its simple-root basis, the matrices

$$s_i(e_i) = -e_i, \qquad s_i(e_j) = e_j + 2\cos(\pi/m_{ij})e_i \quad (j \neq i).$$

The usual convention for $m_{ij} = \infty$ replaces the coefficient by 2. We retain the faithfulness theorem for this representation. Each finite-label coefficient is an algebraic integer: $2\cos(\pi/m) = \zeta_{2m} + \zeta_{2m}^{-1}$. There are only finitely many labels, so all coefficients lie in the ring of integers of a single cyclotomic number field. The generators are involutions and have determinant -1. Thus both the matrices and their inverses have integral coefficients, and Lemma 2 applies. This proves the particular residual-finiteness assertion used below without invoking a normal-subgroup classification.

Corollary 3. *Let (W, S) be an infinite Coxeter system of finite rank at least three, with every maximal standard parabolic finite. For every positive integer M there is a finite regular hypertope with this diagram, with universal spherical proper residues, and with at least M chambers.*

Proof. Choose M distinct elements $w_1, \ldots, w_M$ of W and enlarge the finite set $\mathcal{F}$ by the nonidentity elements $w_a^{-1}w_b$ for $a \neq b$. Apply the faithful representation and Lemma 2 to separate this entire set simultaneously. The image group Q is finite, and its kernel satisfies Theorem 1. The images of the w_a are distinct, so $|Q| \geq M$. The regular chamber action identifies $|Q|$ with the number of chambers of the resulting hypertope. □

This proof is effective in an elementary sense. The finite parabolics can be enumerated in the exact representation, their triple products formed, and a nonzero matrix entry selected for every forbidden element. The norm calculation then supplies a prime to avoid. The finite matrix image can itself be enumerated. These steps need not be computationally practical; no small-order bound is asserted. Taking unbounded M gives infinitely many pairwise nonisomorphic finite hypertopes, rather than just many maps onto the same object.

3 Source consequences and attribution

Section 6 of the fixed manuscript [1] recalls infinitely many finite maps and hypermaps for the compact hyperbolic triangle diagrams. It also recalls, from the classical polytope literature, infinitely many finite locally spherical polytopes for the string diagrams in ranks four and five. Corollary 3 supplies one proof for these assertions. Indeed a compact hyperbolic Coxeter simplex has finite proper parabolics, and its reflection group is infinite. The same argument applies to the nonstring compact hyperbolic diagrams in the source's Table 2. We retain that diagram classification; the corollary does not provide a new proof that the list

is exhaustive. For a string diagram the acquired string coset correspondence identifies the resulting hypertope with a regular abstract polytope.

The conclusion uses actual universal spherical residues. The source also uses "spherical" for the weaker finite-diagram condition, which may permit projective residues. Our existence result satisfies the stronger condition. It cannot be used to rule out smaller examples with projective residues. Nor does it identify a quotient group with any particular extension listed in Table 3, prove a minimum there, or verify a quoted exceptional construction. Those remain separate finite or attributed inputs.

The finite existence results for string diagrams are prior work, as are residual finiteness of Coxeter groups and the standard Coxeter complex. The contribution of this component is the explicit triple-product argument linking a finite separating set to full flag lifting and residue incidence, followed by the common application. No claim of historical priority is made for this argument. The earlier project note *Finite quotients with prescribed parabolic intersections* treats different hypotheses involving separability and endomorphisms; none of its unproved filling assertions is used here. This component makes no claim to resolve a source conjecture, complete the entire L033 manuscript, provide a Lean formalization, or establish a numerical compression factor.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*, fixed sixteen-page revised manuscript, catalogue L033, Section 6 and Table 2. Its classical existence citations and the finite table are distinguished above.

[2] Human Compression project, *Residue stabilizers and distinguished generators*; *Kernels and intersections in quotients of hypertopes*. The present proof retains the universal Coxeter complex and its standard intersection property, and proves the quotient interfaces explicitly.